A FINITELY GENERATED ALGEBRAIC EXTENSION IS FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The engine behind every finiteness reduction in the chapter, including the algebraic-elements subextension lemma (b7 card L48374).

Editorial note. The label lemma-algebraic-finitely-generated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-field-extension-generated-by-one-element (source lemma, worklist row 16; set up by the observation immediately preceding at source line 907): an extension generated by one algebraic element is finite
- Chapter lemma-multiplicativity-degrees (delivered card L48338): degrees multiply in towers

Source reference. Original reference 09G9: sources/stacks-fields/pinned-original-source.tex, source lines 759–759. Prerequisite pointer invoked at 924; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field, and let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be elements of some extension field such that each α_i is algebraic over k. Then the extension $k(\alpha_1, \ldots, \alpha_n)/k$ is finite. That is, a finitely generated algebraic extension is finite.*

Proof. Indeed, each extension $k(\alpha_1, \ldots, \alpha_{i+1})/k(\alpha_1, \ldots, \alpha_i)$ is generated by one element and algebraic, hence finite. By multiplicativity of degree (Lemma 09G9) we obtain the result. □

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